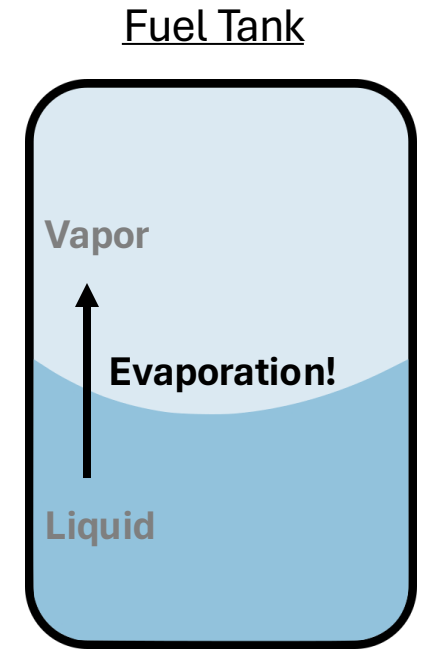
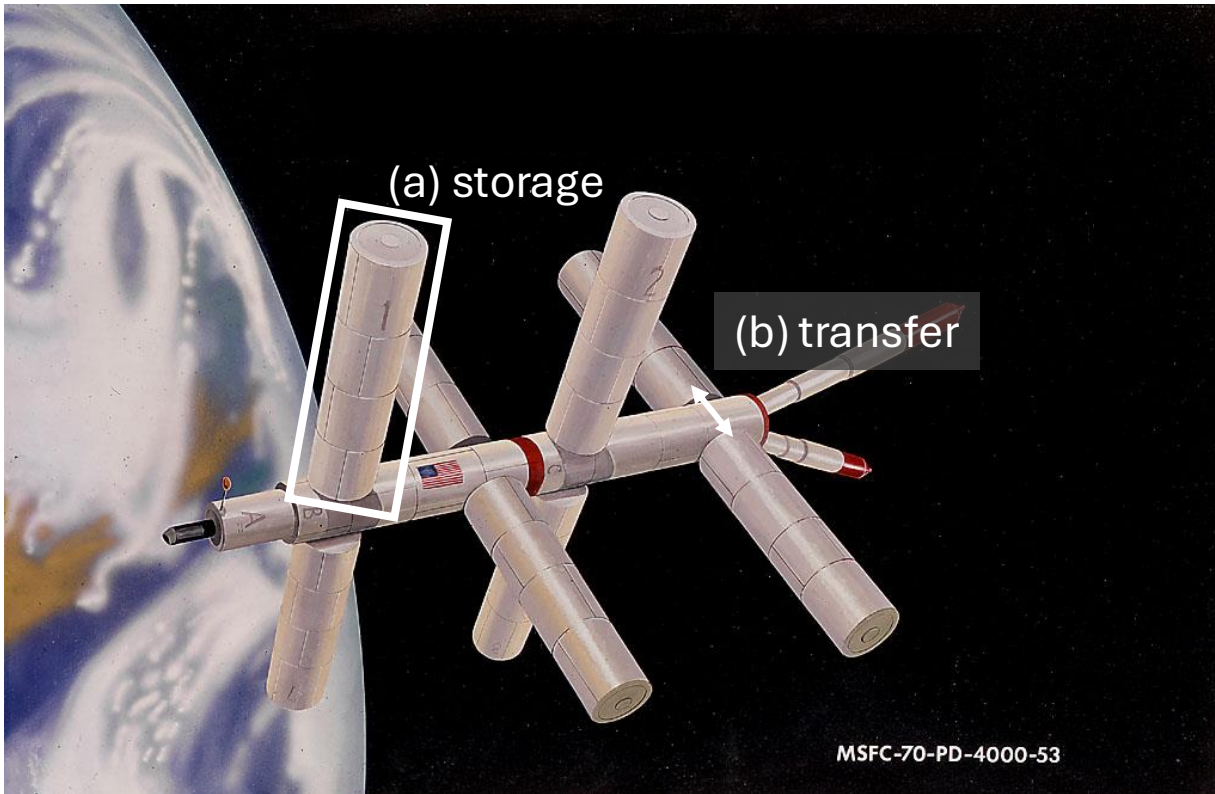


A multiscale approach to modelling evaporation in cryogenic fuels

Saaras Pakanati, Ayaaz Yasin, Kishan Bellur

Motivation and Problem Statement

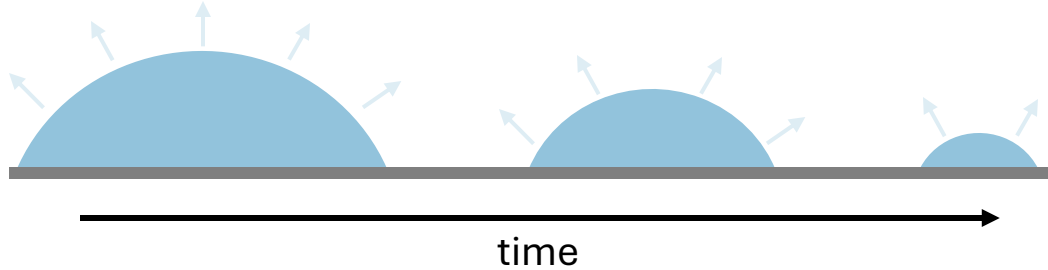
“Developing architecture on the Moon and Mars to enable explorers to go — and stay — requires safe storage of propellants.”
- NASA



Excessive Vapor causes lost margins and, in the worst case, tank pressurization.

Evaporation

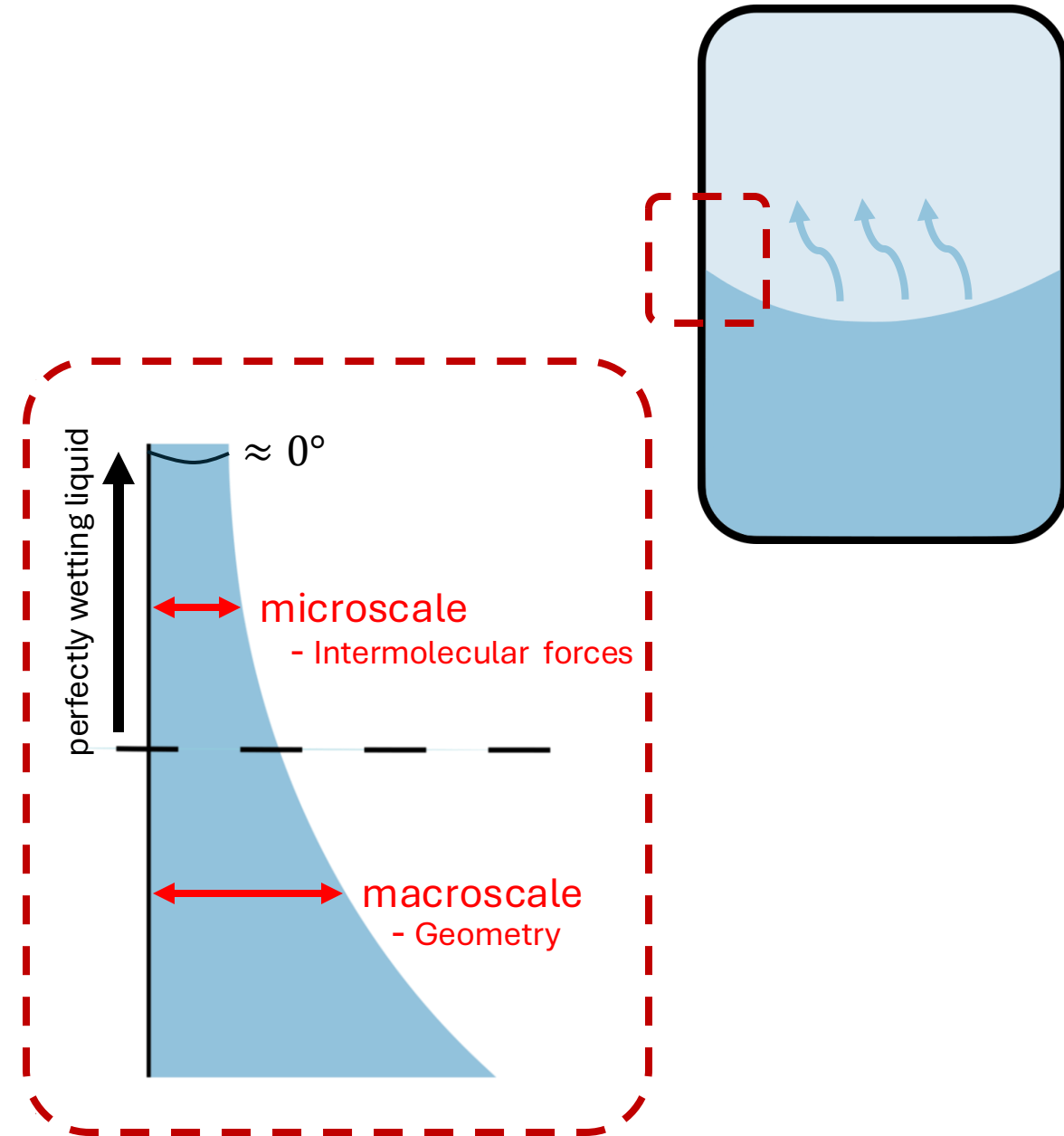
“Evaporation is the process by which liquid changes into its vapor state at temperatures below its boiling point.”



Complications associated with Hydrogen:

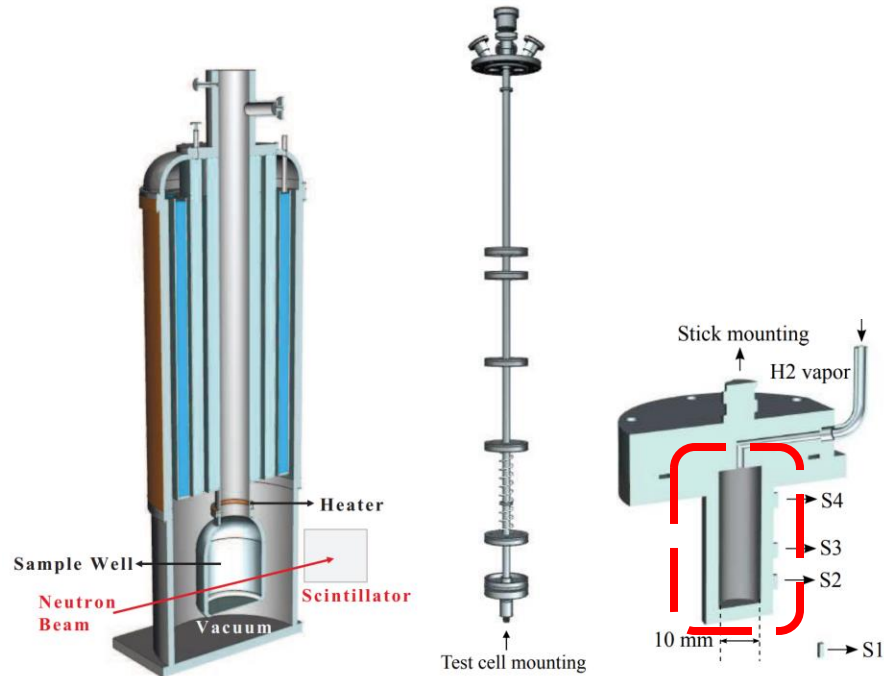
- It is a cryogenic fluid (sat. Temp. 21 K).
- It is a perfectly wetting liquid.

A physics-based evaluation of an experimental dataset is necessary to understand hydrogen evaporation.

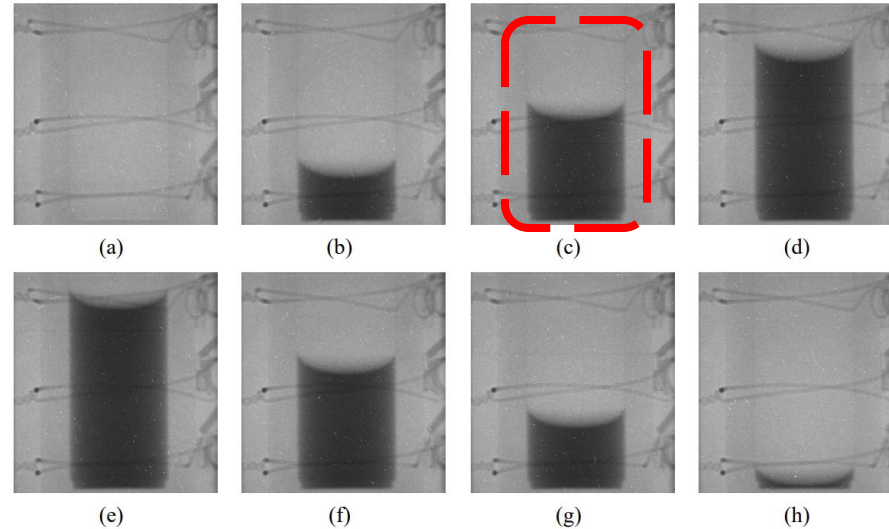


The experimental dataset

A team at Michigan Technological University and NIST conducted the cryo-neutron experiments.



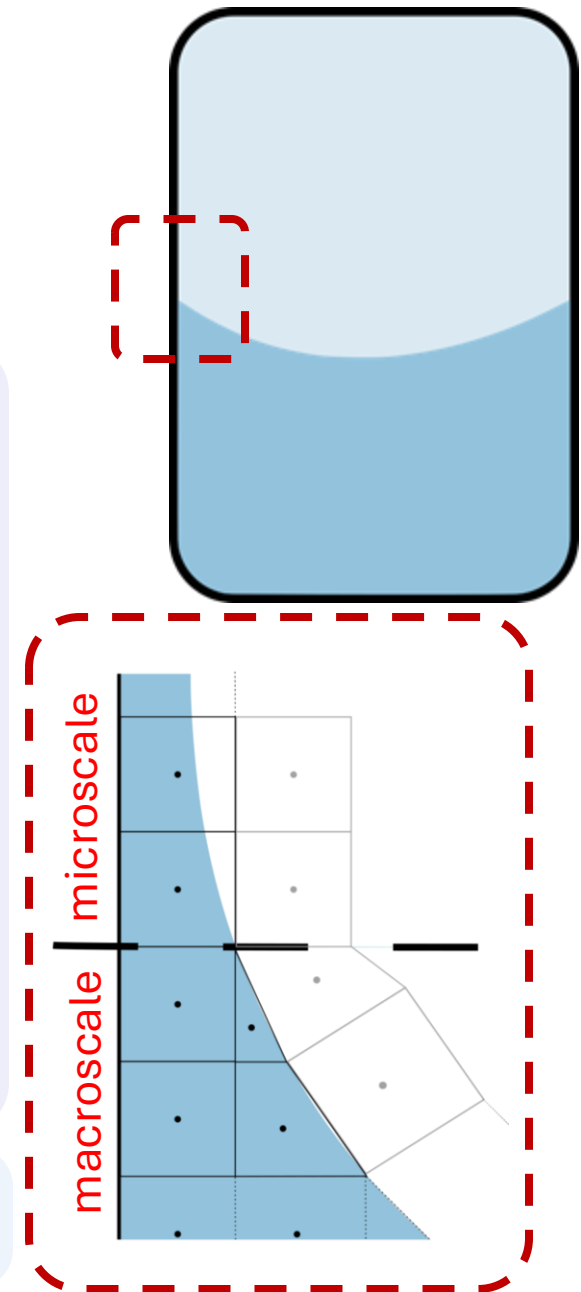
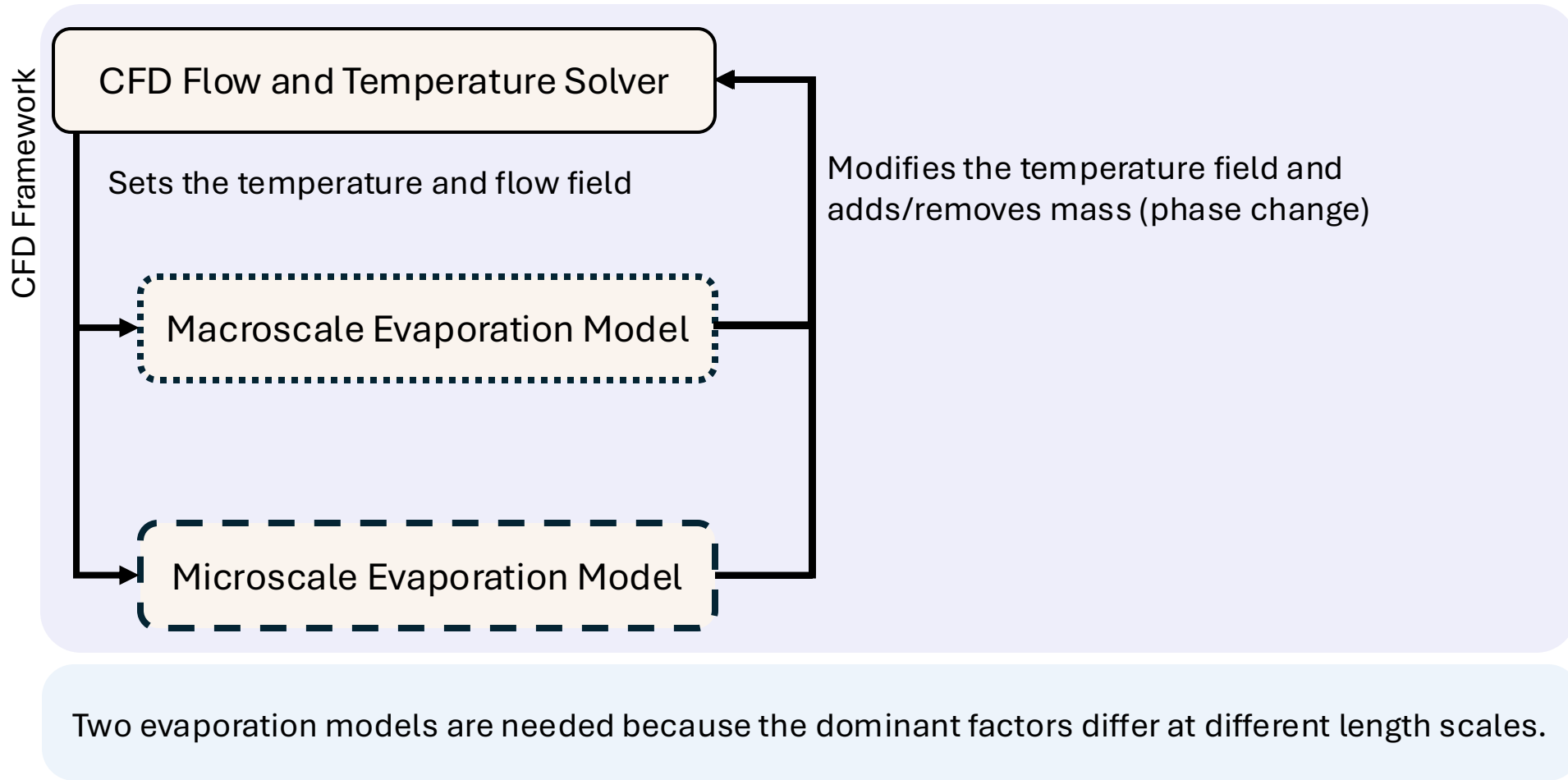
Evaporating and Condensing Hydrogen



This experiment provides evaporation rates of hydrogen in Aluminum and Stainless-Steel containers.

Our study - Overview

Our goal is to simulate hydrogen fuel evaporation using physics-based custom models and commercially available CFD software.



Macroscale evaporation model

A Kinetic Theory of Gases-based approach is used.

Macroscale Evaporation Model

Compute evaporated mass

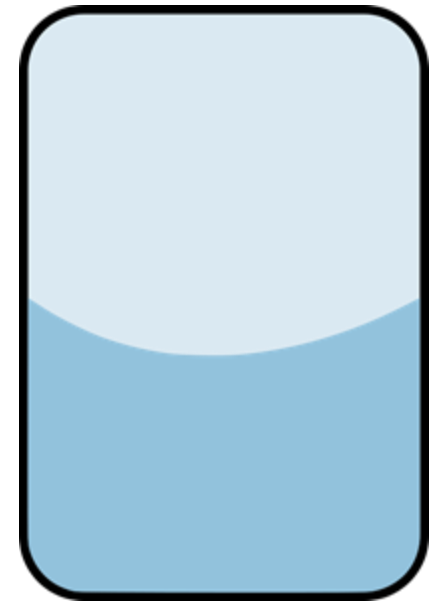
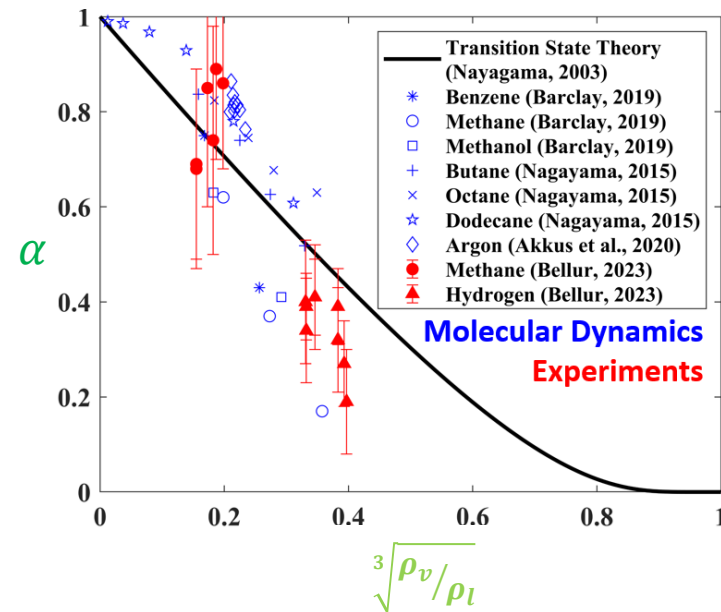
- Compute accommodation coefficient (α_c)
- Compute reduced vapor temperature (T_v^*)

Move mass and heat due to evaporation

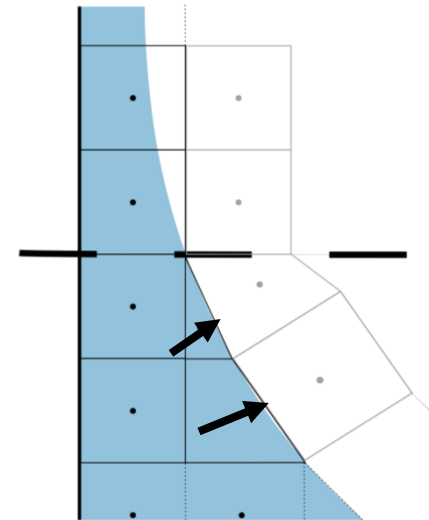
$$S_m = \frac{1}{V_{\text{cell}}} \iint_S m'' dA \quad S_{\text{LH}} = Q_{\text{LH}}'' \times \frac{A_{\text{cell}}}{V_{\text{cell}}}$$

$$\dot{m}'' = \frac{2\alpha_c}{2 - \alpha_c} \sqrt{\frac{M}{2\pi RT_v}} p_v \left[\beta W \sqrt{\frac{T_v^*}{T_i}} - 1 \right]$$

$$\alpha_c = \left(1 - \sqrt[3]{\rho_v/\rho_l} \right) \exp \left(\frac{-\sqrt[3]{\rho_v/\rho_l}}{2(1 - \sqrt[3]{\rho_v/\rho_l})} \right)$$



$$T_v^* = T_v(1 - \gamma)$$



Microscale evaporation model

Microscale Evaporation Model

Compute evaporated mass

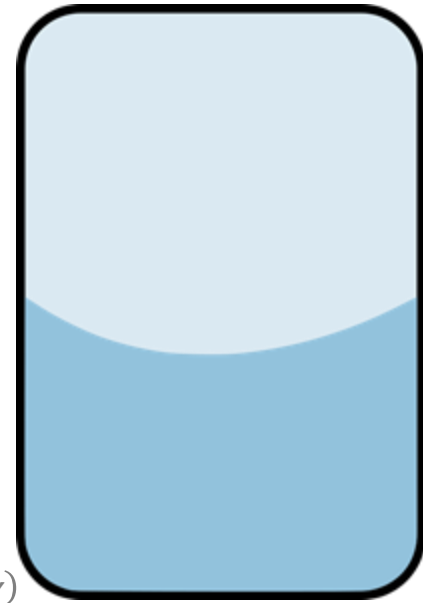
- Compute accommodation coefficient (α_c)
- Compute reduced vapor temperature (T_v^*)

+

More computations...

$$\dot{m}'' = \frac{2\alpha_c}{2 - \alpha_c} \sqrt{\frac{M}{2\pi \bar{R} T_v}} p_v \left[\beta W \sqrt{\frac{T_v^*}{T_i}} - 1 \right]$$

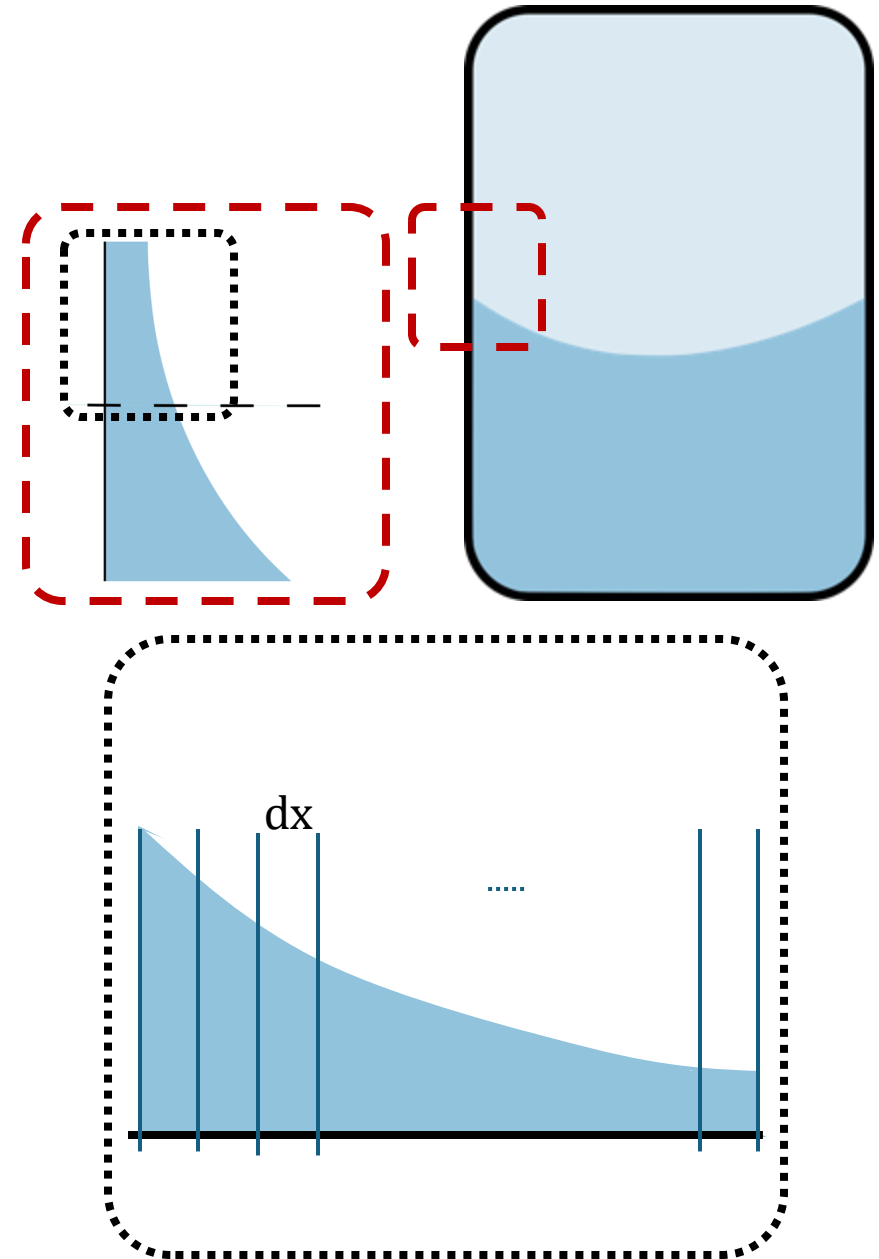
$$\alpha_c = \left(1 - \sqrt[3]{\rho_v / \rho_l} \right) \exp \left(\frac{-\sqrt[3]{\rho_v / \rho_l}}{2(1 - \sqrt[3]{\rho_v / \rho_l})} \right) \quad T_v^* = T_v(1 - \gamma)$$



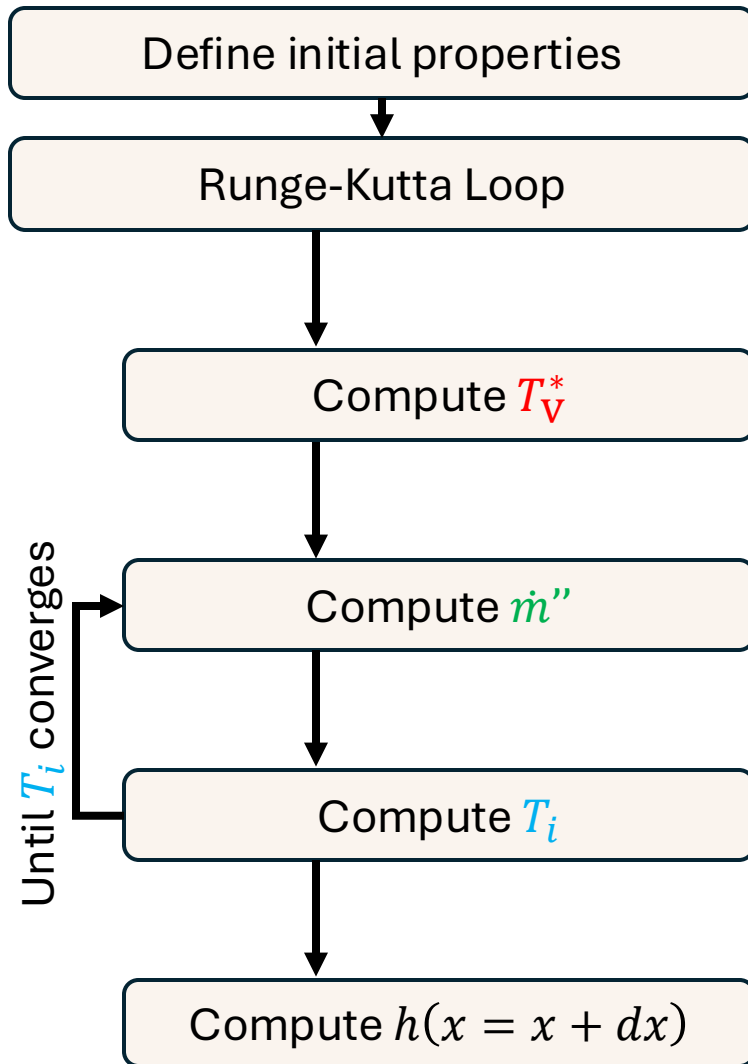
Define Initial Properties

Define initial properties

- Initial height, h and its derivatives
- Wall temperature, T_w
- Material properties as functions of temperature
- Reduction factor γ



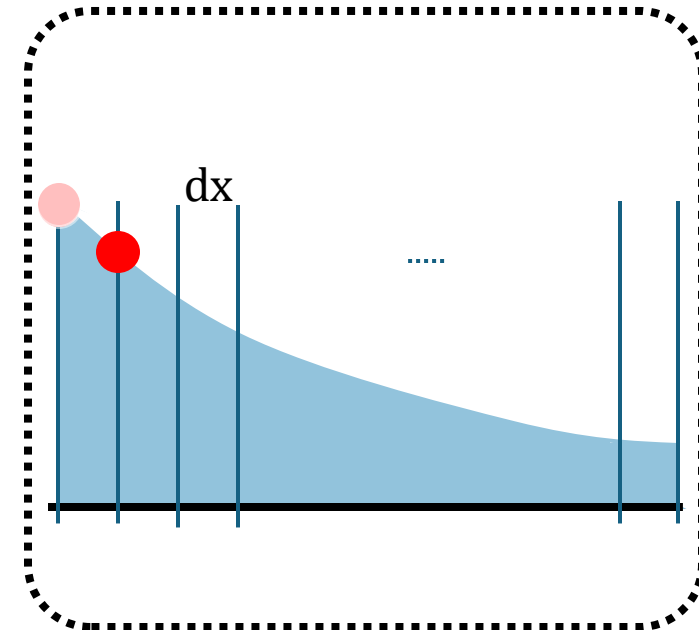
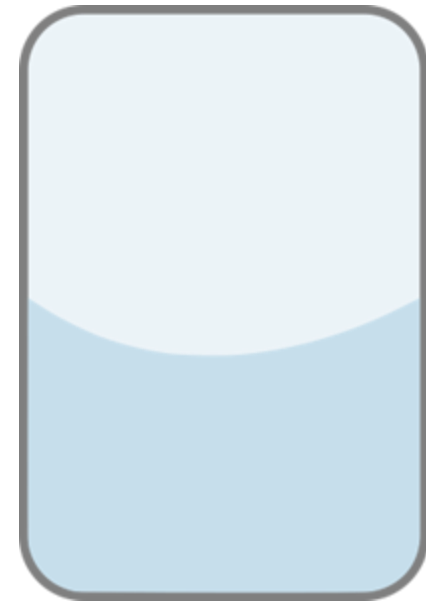
Iterate through the film – RK4 (Part 1)



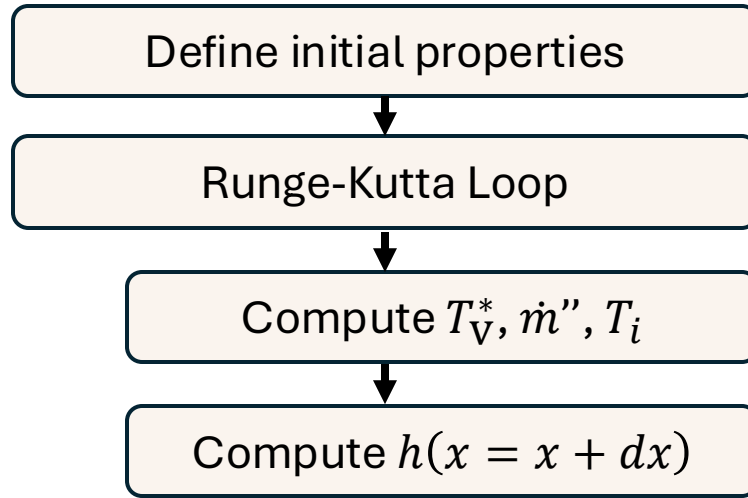
$$T_v^* = T_v(1 - \gamma)$$

$$\dot{m}'' = \frac{2\alpha_c}{2 - \alpha_c} \sqrt{\frac{M}{2\pi\bar{R}T_v}} p_v \left[\beta W \sqrt{\frac{T_v^*}{T_i}} - 1 \right]$$

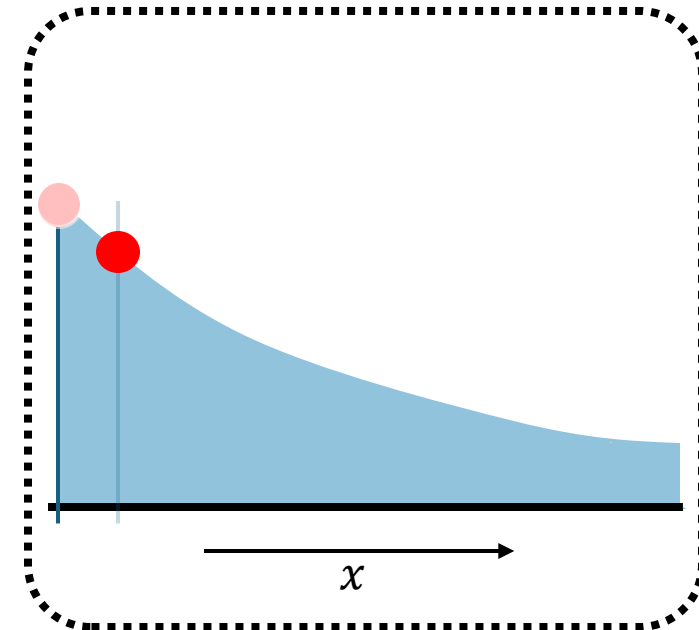
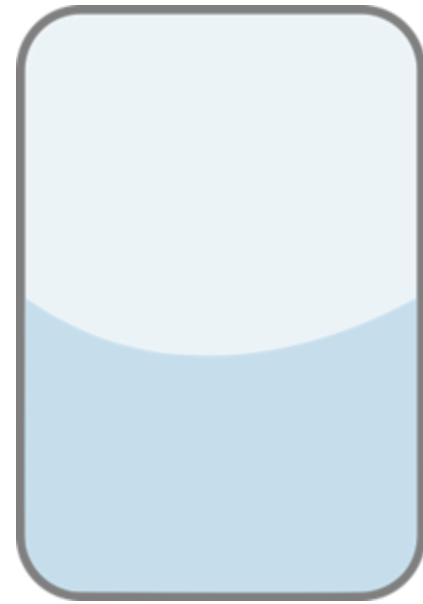
$$T_i = T_w - \dot{m}'' \frac{h_{fg}}{k} (R - h) \ln\left(\frac{R}{R - h}\right)$$



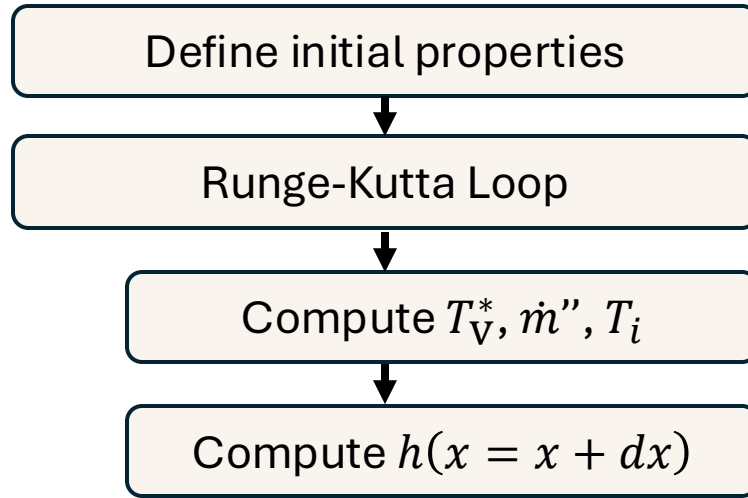
Iterate through the film – RK4 (Part 2)



$$h_{xxx} - \frac{3h_x h_{xx}^2}{1 + h_x^2} - \frac{h_x h_{xx}}{R - h} + \frac{h_x(1 + h_x^2)}{(R - h)^2} + \left(\frac{1 + h_x^2}{R - h} + h_{xx} \right) \frac{1}{\sigma} \frac{d\sigma}{dx} + \frac{(1 + h_x^2)^{3/2}}{\sigma} \left(\frac{dp_l}{dx} + \frac{d\Pi}{dx} \right) = 0$$

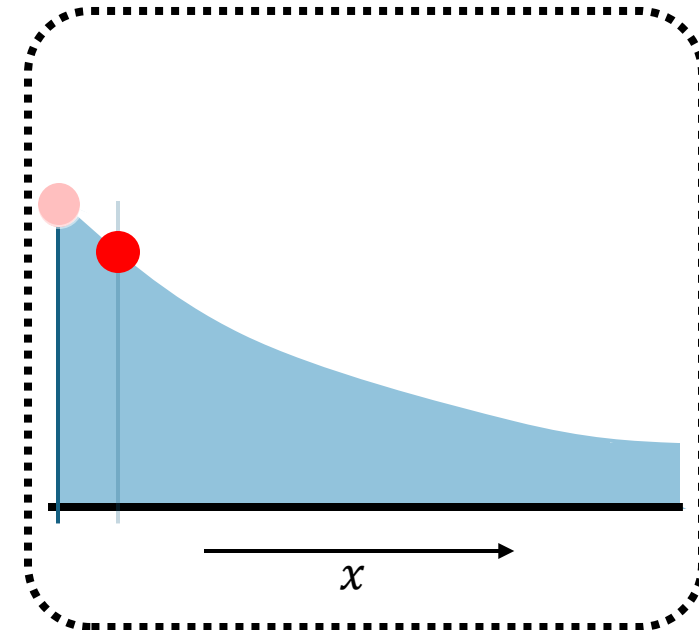
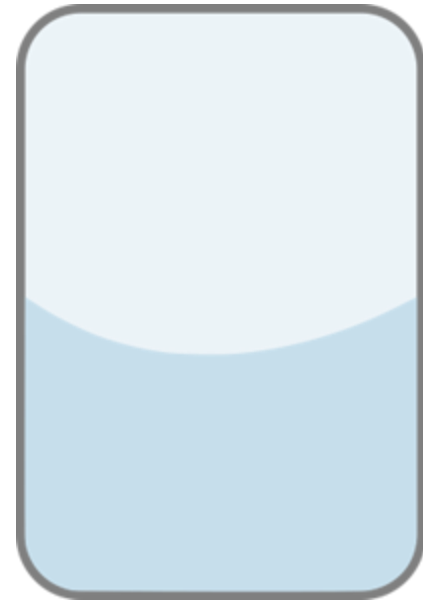


Iterate through the film – RK4 (Part 2)

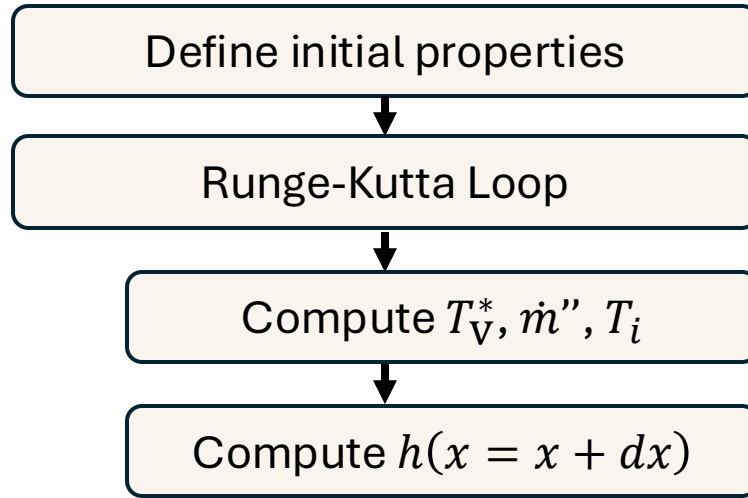


$$h_{xxx} - \frac{3h_x h_{xx}^2}{1 + h_x^2} - \frac{h_x h_{xx}}{R - h} + \frac{h_x(1 + h_x^2)}{(R - h)^2} + \left(\frac{1 + h_x^2}{R - h} + h_{xx} \right) \frac{1}{\sigma} \frac{d\sigma}{dx} + \frac{(1 + h_x^2)^{3/2}}{\sigma} \left(\frac{dp_l}{dx} + \frac{d\Pi}{dx} \right) = 0$$

$$\frac{d\sigma}{dx} = \sigma_T \frac{dT_i}{dx}$$

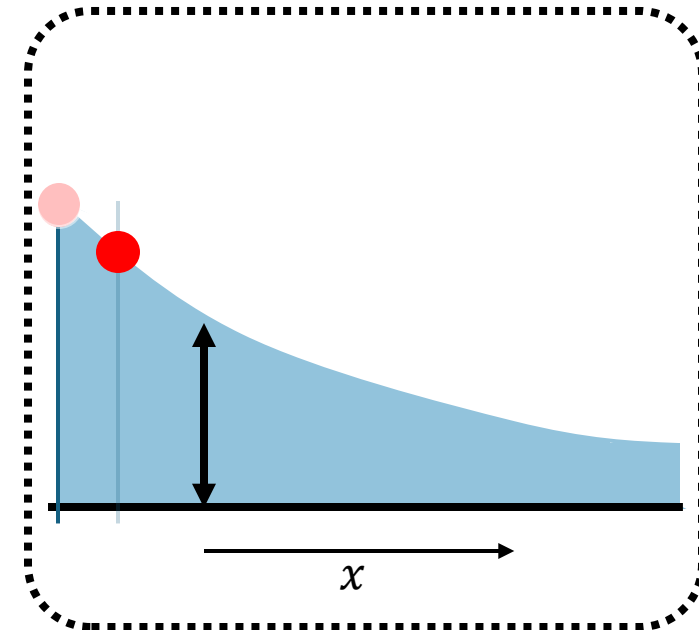
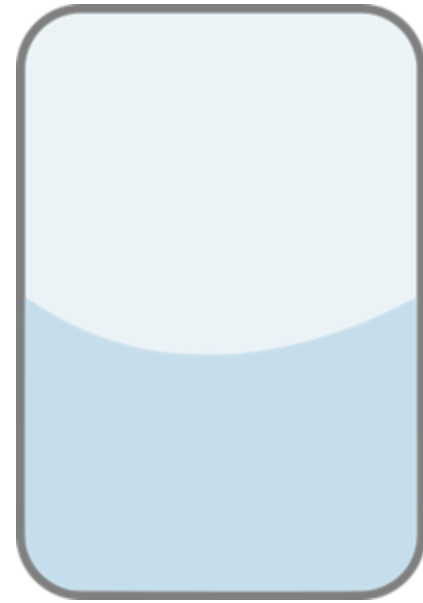


Iterate through the film – RK4 (Part 3)

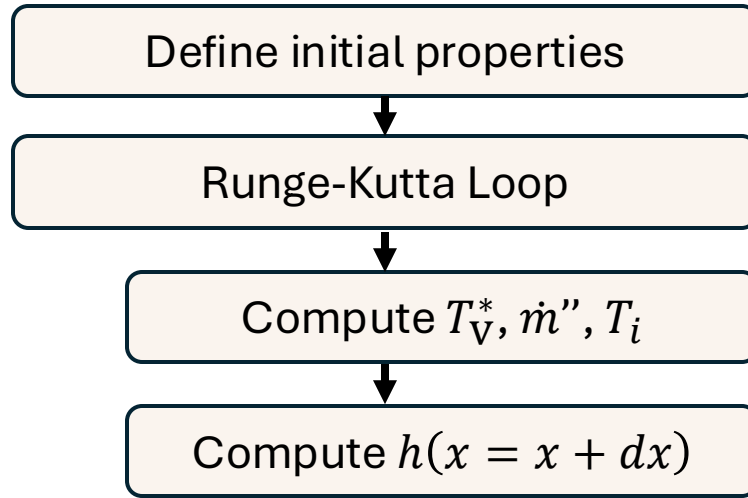


$$h_{xxx} - \frac{3h_x h_{xx}^2}{1 + h_x^2} - \frac{h_x h_{xx}}{R - h} + \frac{h_x(1 + h_x^2)}{(R - h)^2} + \left(\frac{1 + h_x^2}{R - h} + h_{xx} \right) \frac{1}{\sigma} \frac{d\sigma}{dx} + \frac{(1 + h_x^2)^{3/2}}{\sigma} \left(\frac{dp_l}{dx} + \frac{d\Pi}{dx} \right) = 0$$

$$\frac{d\sigma}{dx} = \sigma_T \frac{dT_i}{dx} \quad \Pi = \frac{\mathcal{A}}{h^3}$$



Iterate through the film – RK4 (Part 4)

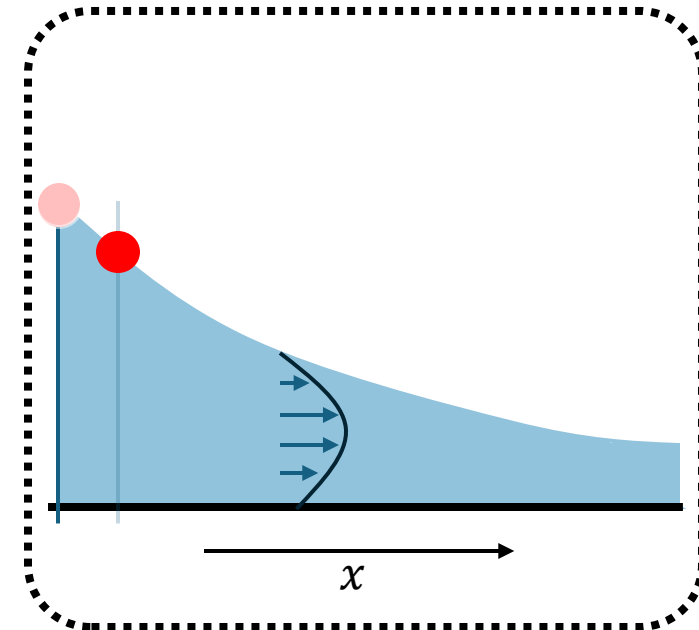
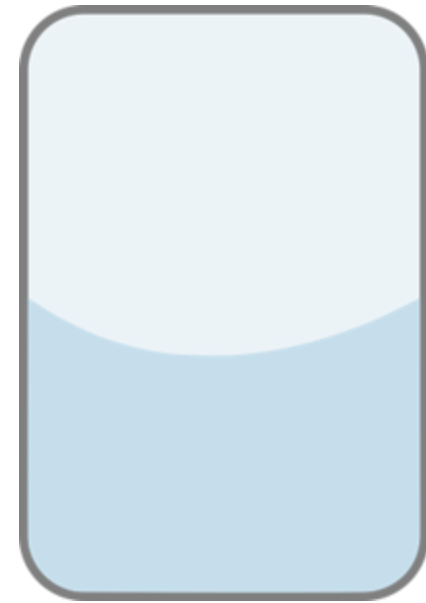


$$h_{xxx} - \frac{3h_x h_{xx}^2}{1 + h_x^2} - \frac{h_x h_{xx}}{R - h} + \frac{h_x(1 + h_x^2)}{(R - h)^2} + \left(\frac{1 + h_x^2}{R - h} + h_{xx} \right) \frac{1}{\sigma} \frac{d\sigma}{dx} + \frac{(1 + h_x^2)^{3/2}}{\sigma} \left(\frac{dp_l}{dx} + \frac{d\Pi}{dx} \right) = 0$$

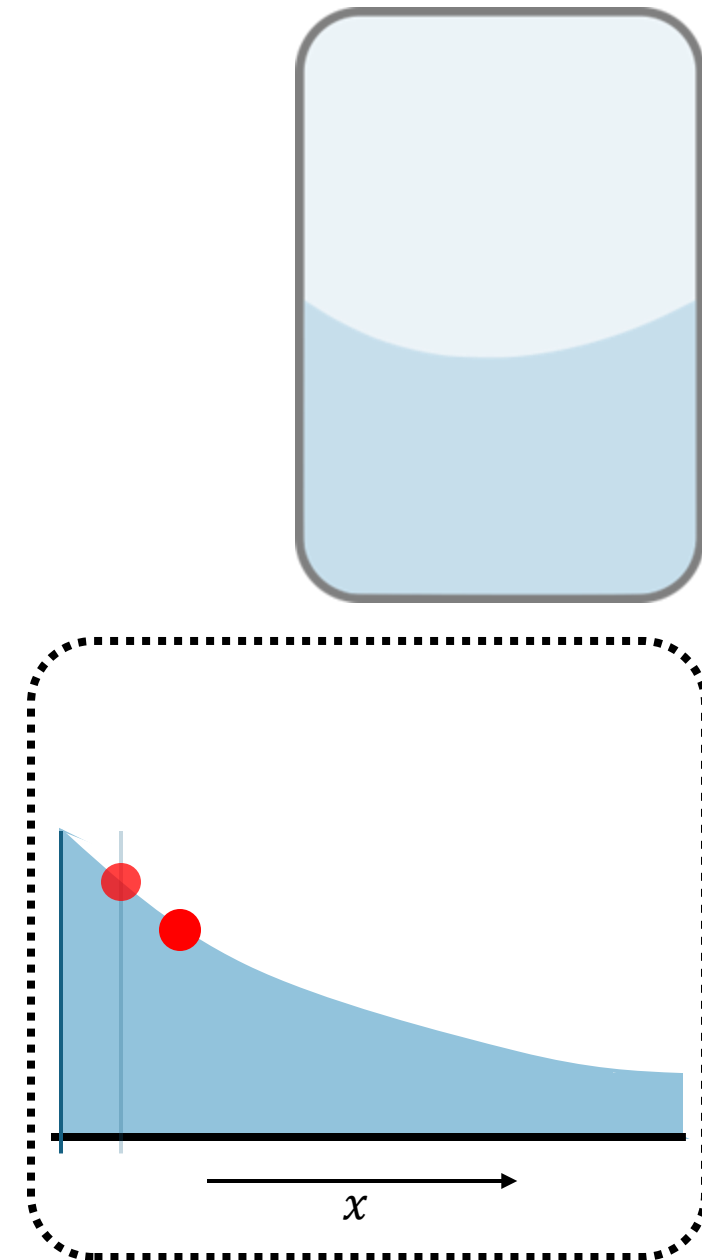
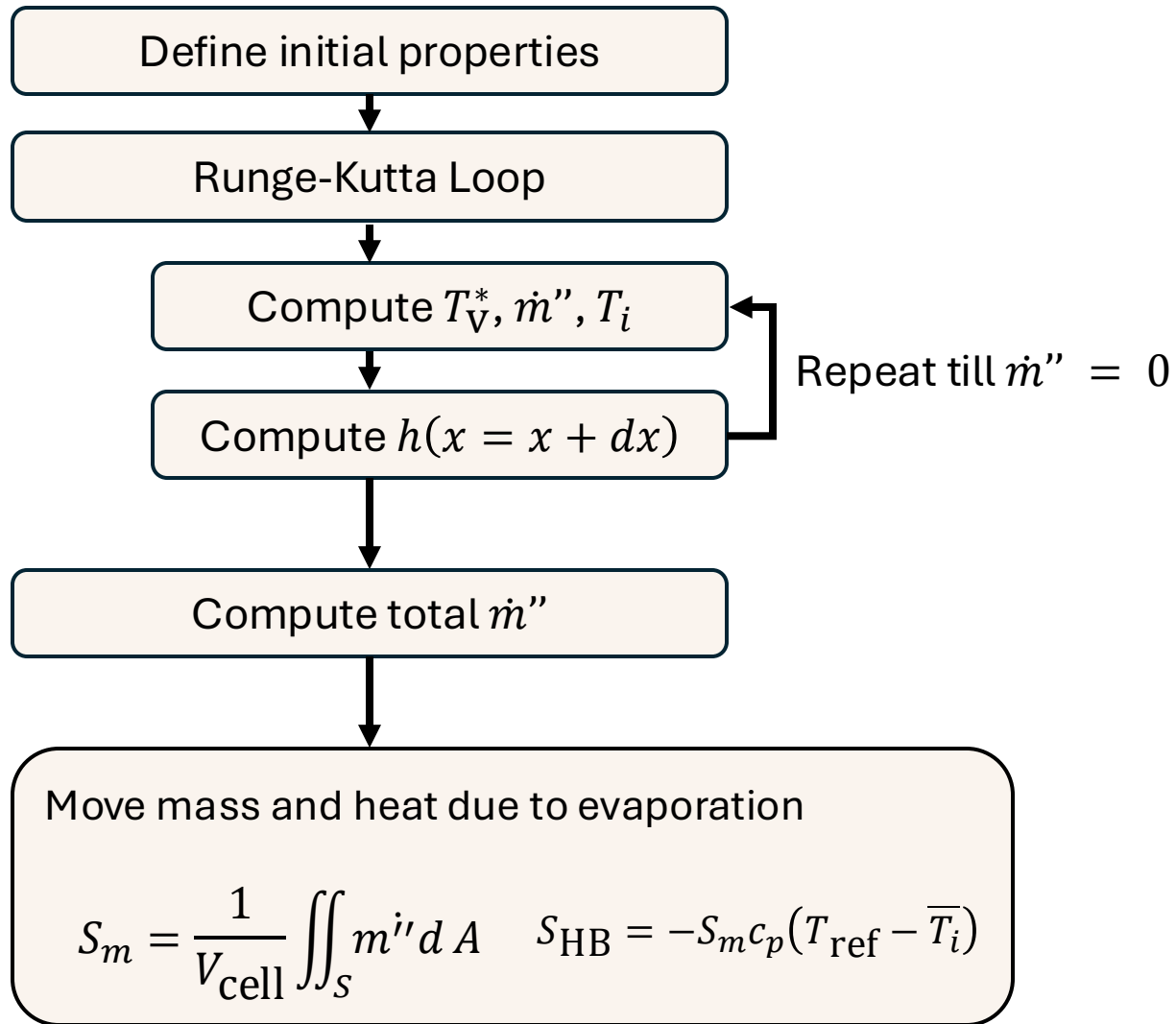
$$\frac{d\sigma}{dx} = \sigma_T \frac{dT_i}{dx}$$

$$\Pi = \frac{\mathcal{A}}{h^3}$$

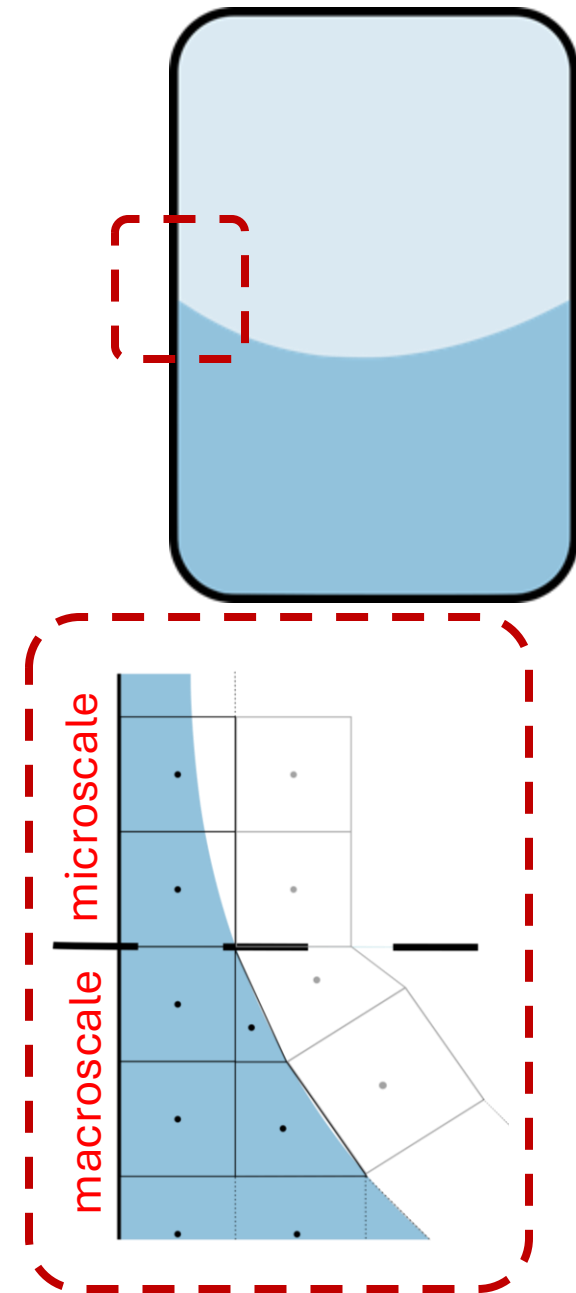
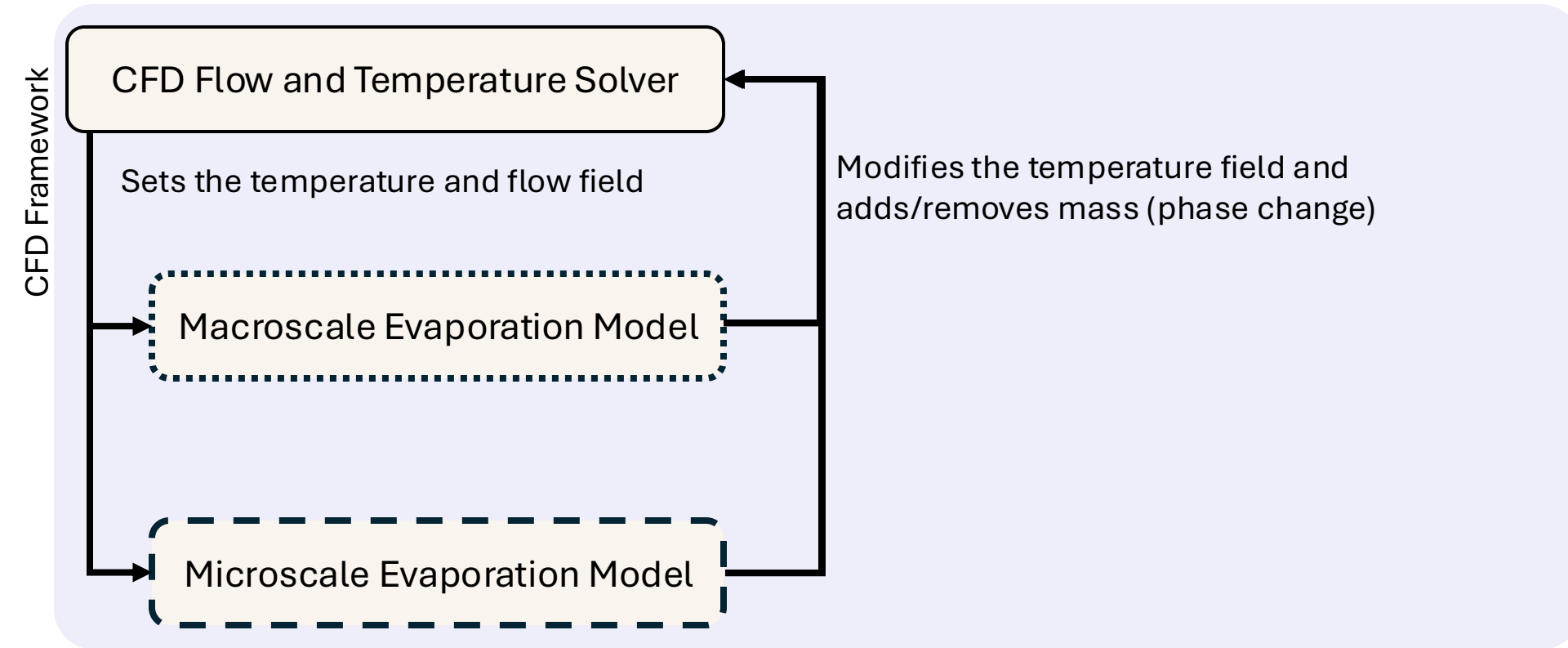
$$\frac{\partial p_l}{\partial x}(x) = \frac{\frac{4\sigma_T}{R - h} \frac{\partial T_i}{\partial x} \left(1 - 2 \ln \frac{R}{R - h} \right) - \frac{8\mu\Gamma}{\pi\rho_l} \frac{1}{h(2R - h)(R - h)^2}}{1 + 2 \left(1 - 2 \ln \frac{R}{R - h} \right) + \frac{R^2}{(R - h)^2}}$$



Integrate microscale model output



Methodology recap



Model validation

3 criteria for validation:

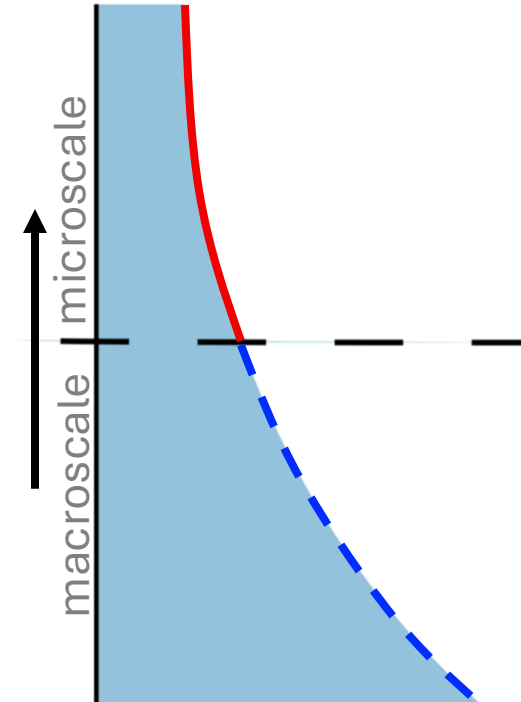
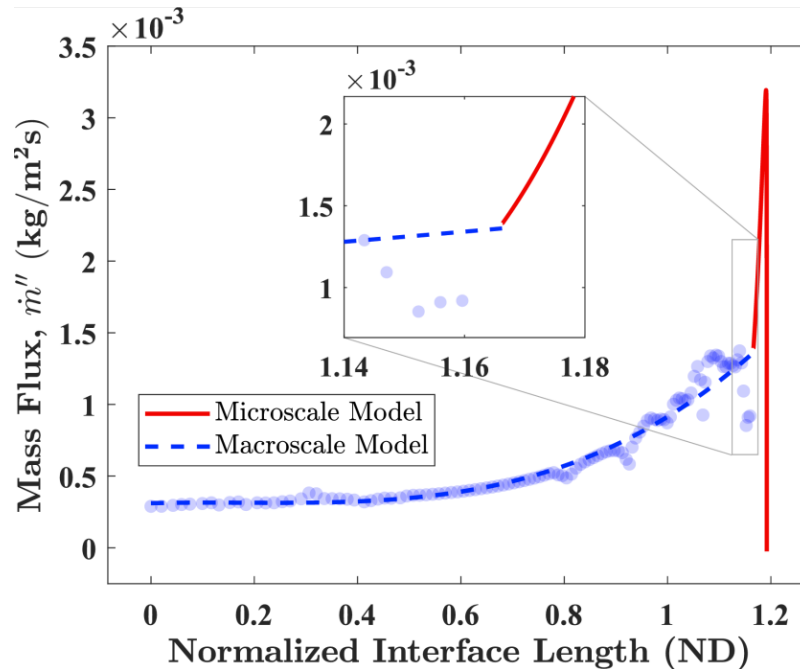
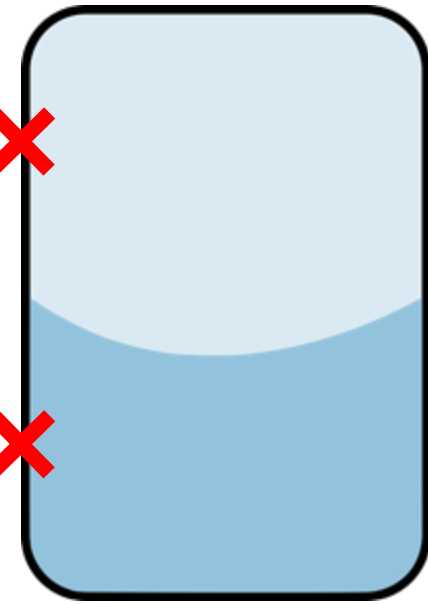
(a) The temperature fields match the experimental data.

(b) The computed evaporation rates match the experiments.

Computed evaporation rates match experimental results within 1%.

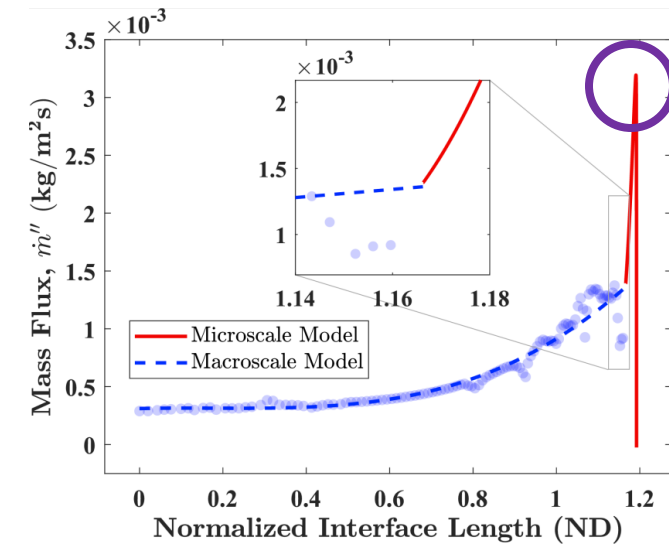
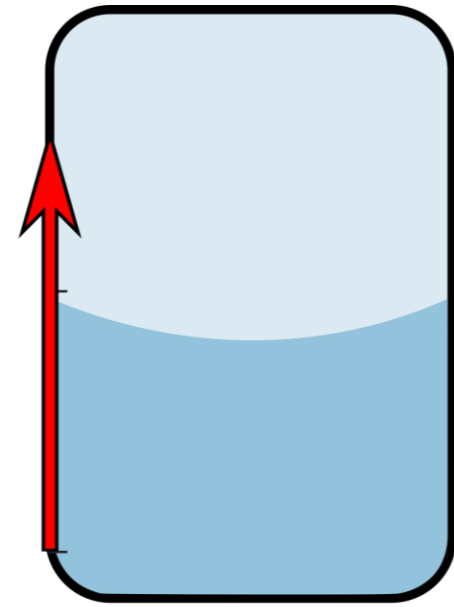
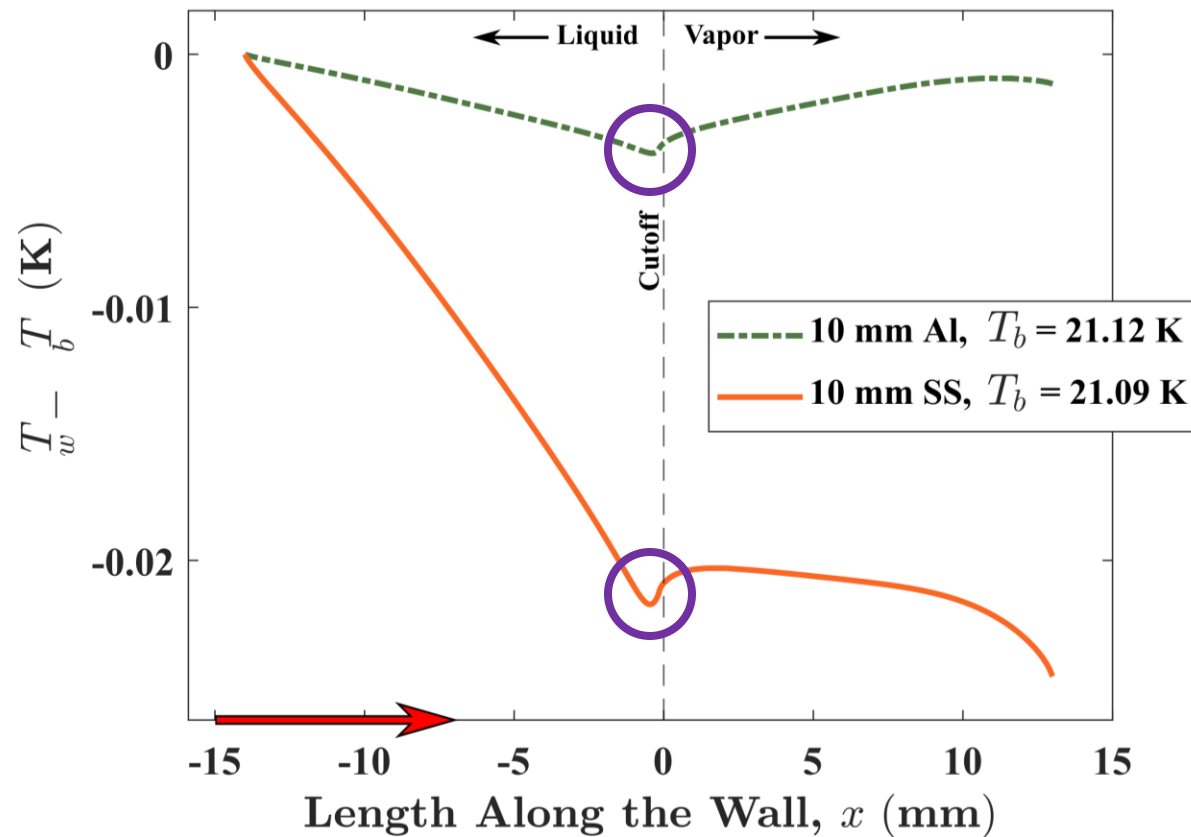
(c) The macroscale and microscale models agree on evaporation values at the “hand off” point.

Thermocouples



Wall temperature variations

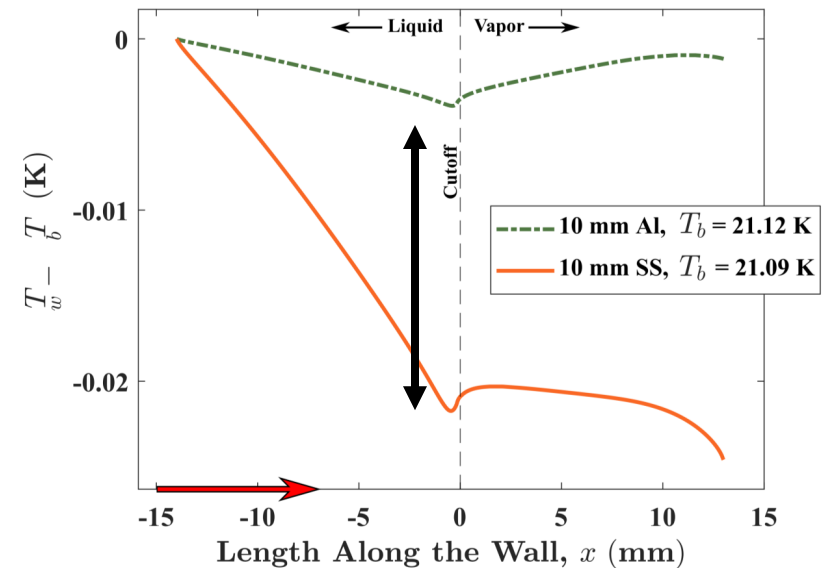
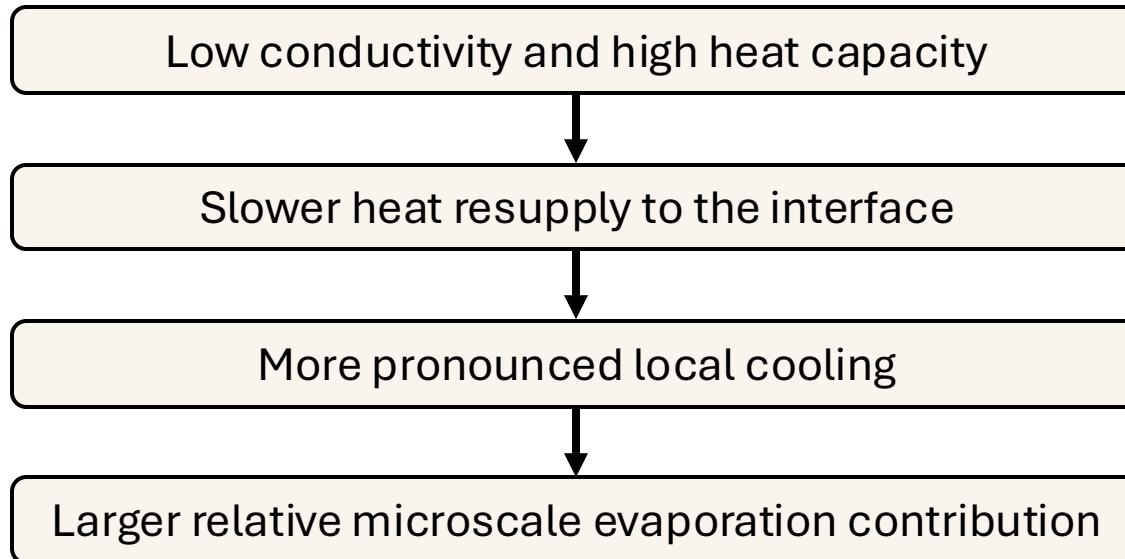
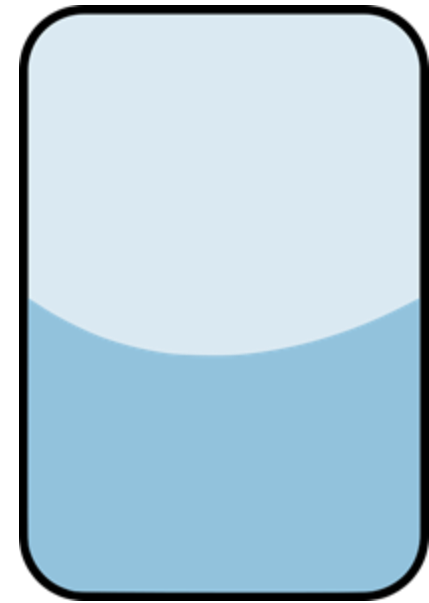
The non-uniform temperature variation along the wall was extracted.



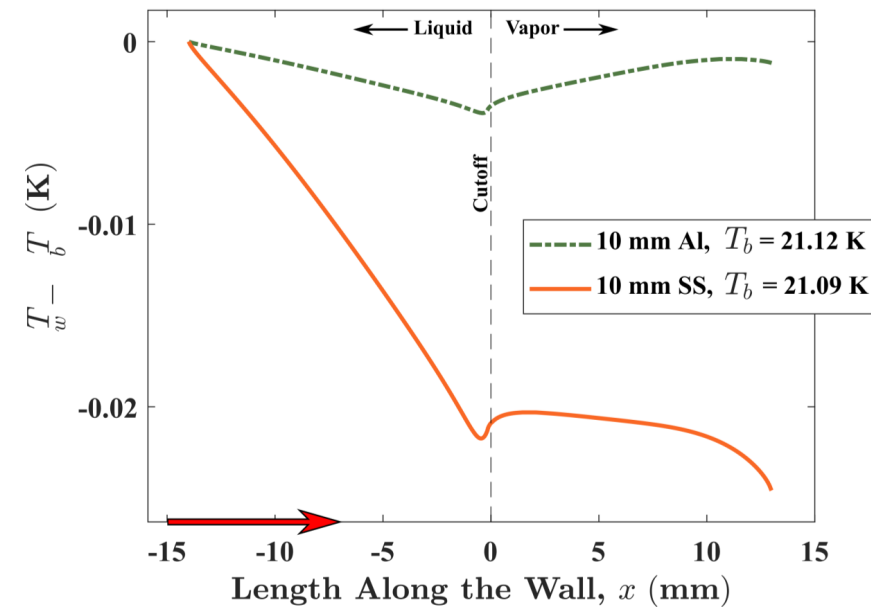
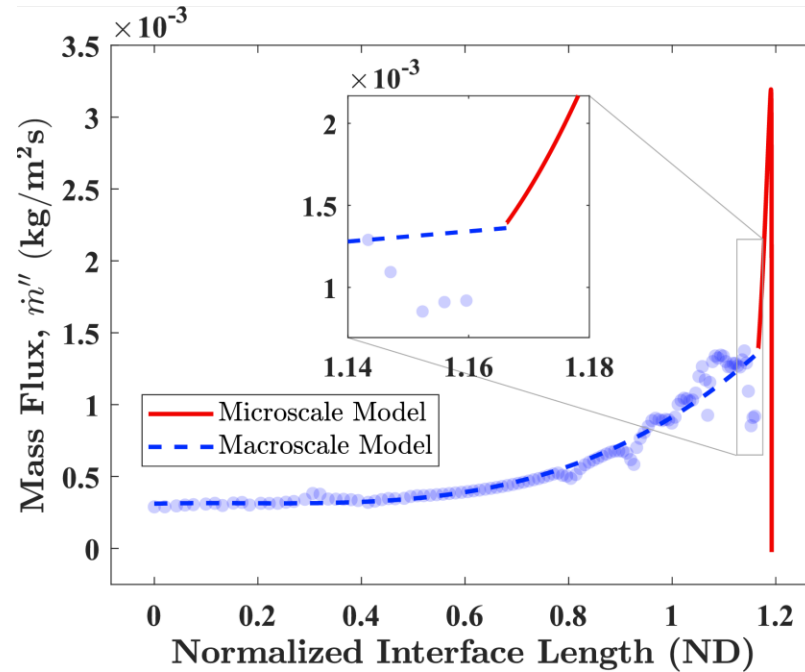
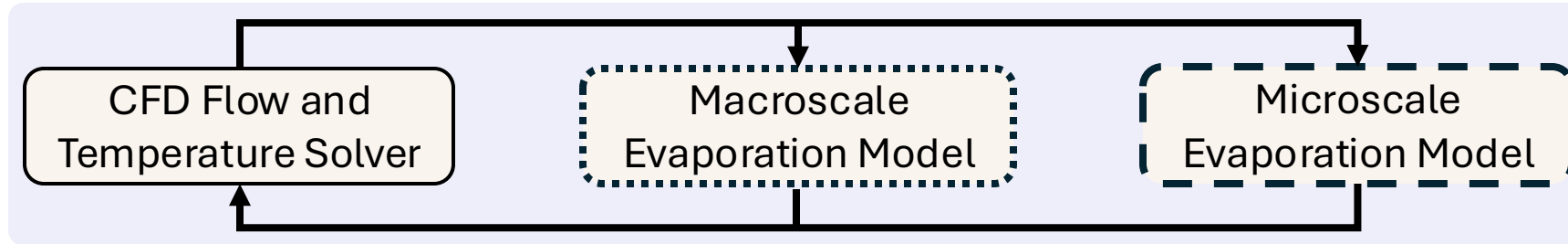
Effect of wall material

Due to the model's iterative nature, the wall material's effect on evaporation dynamics can be extracted. This explains some enhancements beyond those of prior studies.

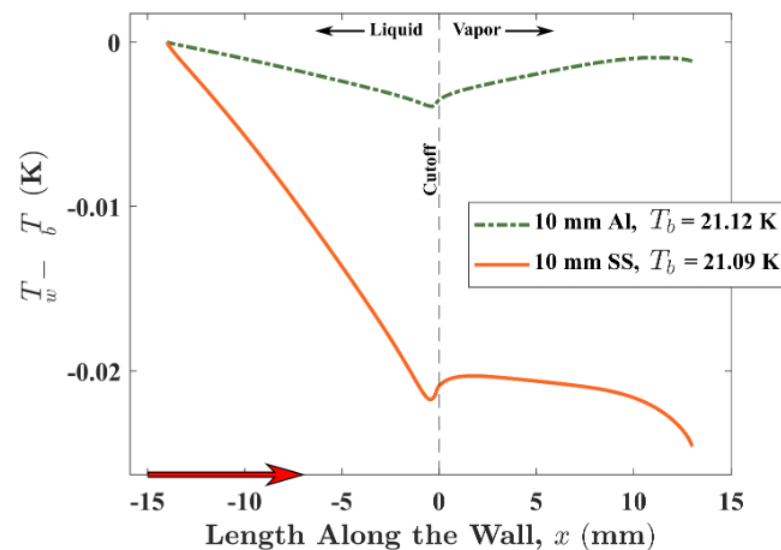
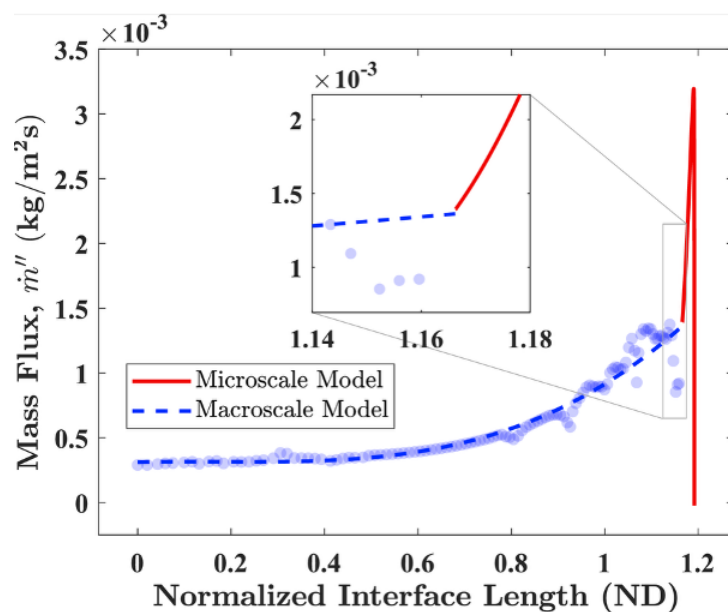
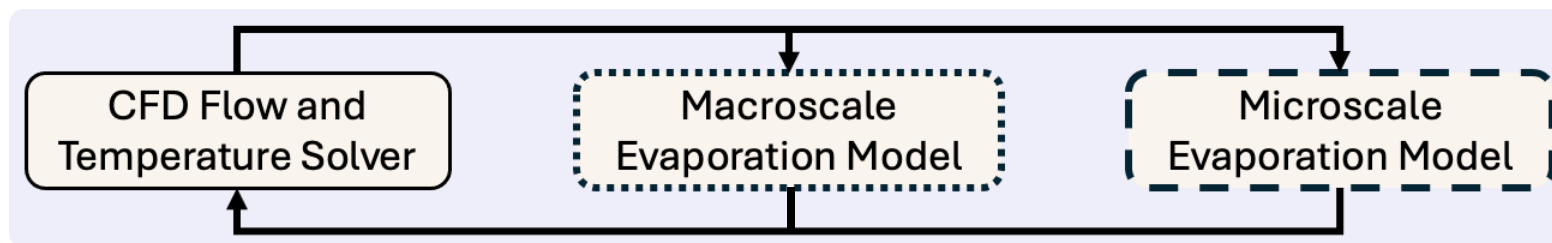
SS 316 exhibits a *higher* microscale contribution in the present study over prior studies. (11% higher)



Conclusion



Thank You!



Collaborators: Ayaz Yasin & Kishan Bellur

Funding Sources:



Appendix

Microscale evaporation model

Microscale Evaporation Model

Compute evaporated mass

- Compute accommodation coefficient (α_c)
- Compute reduced vapor temperature (T_v^*)

+

- Compute film height (h)
- Compute Interfacial Temperature (T_i)

Move mass and heat due to evaporation

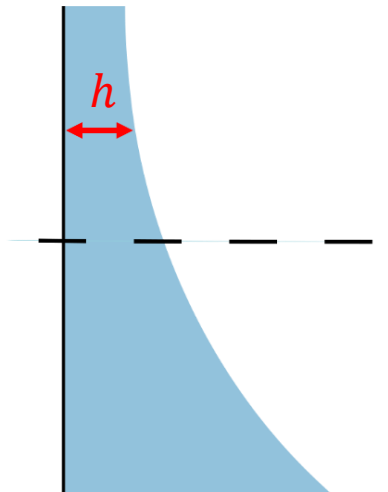
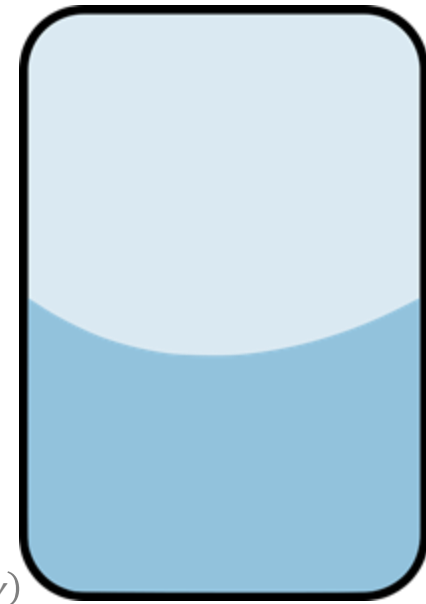
$$S_m = \frac{1}{V_{\text{cell}}} \iint_S \dot{m}'' dA \quad S_{\text{HB}} = -S_m c_p (T_{\text{ref}} - \bar{T}_i)$$

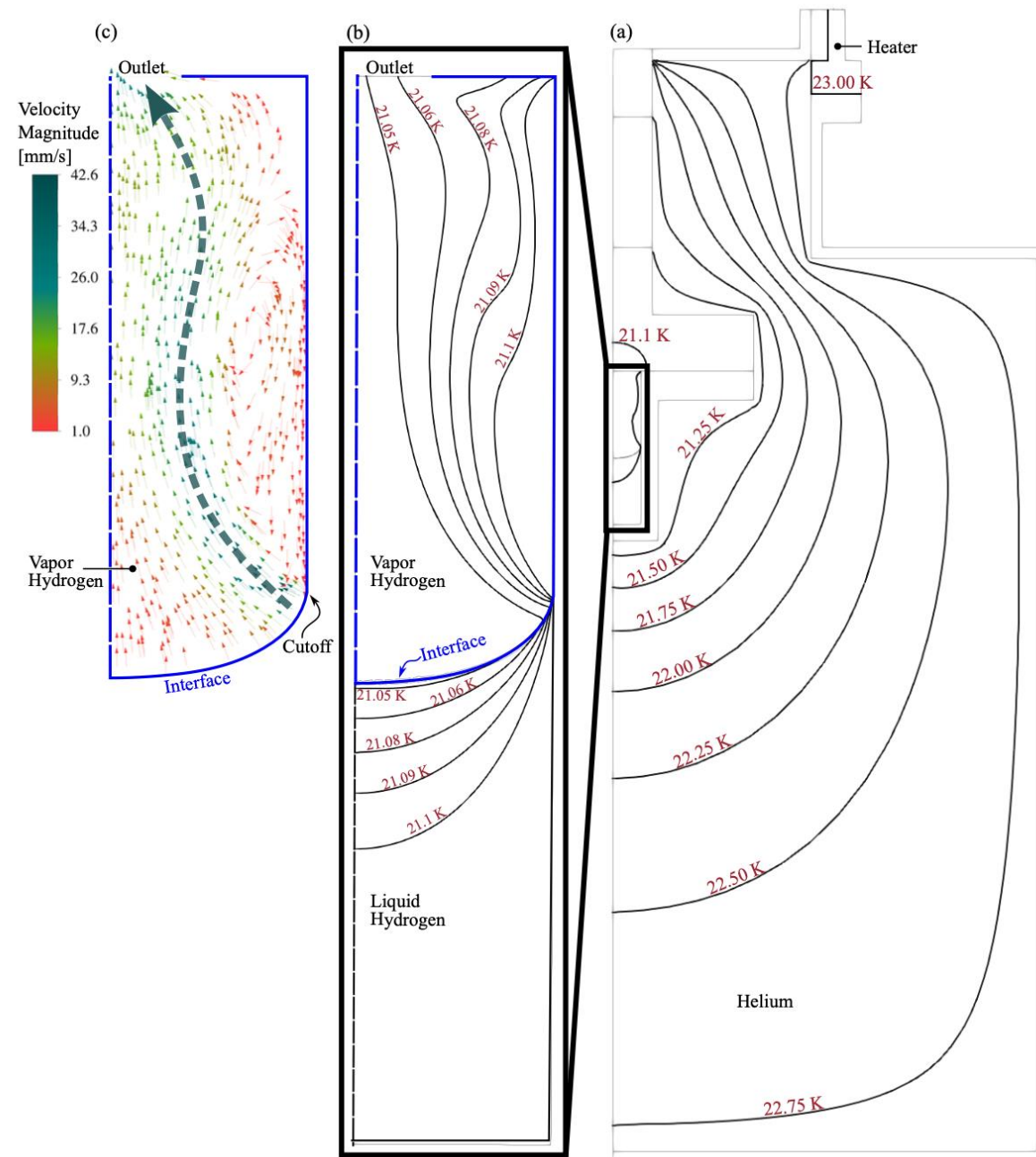
$$\dot{m}'' = \frac{2\alpha_c}{2 - \alpha_c} \sqrt{\frac{M}{2\pi \bar{R} T_v}} p_v \left[\beta W \sqrt{\frac{T_v^*}{T_i}} - 1 \right]$$

$$\alpha_c = \left(1 - \sqrt[3]{\rho_v / \rho_l} \right) \exp \left(\frac{-\sqrt[3]{\rho_v / \rho_l}}{2(1 - \sqrt[3]{\rho_v / \rho_l})} \right) \quad T_v^* = T_v (1 - \gamma)$$

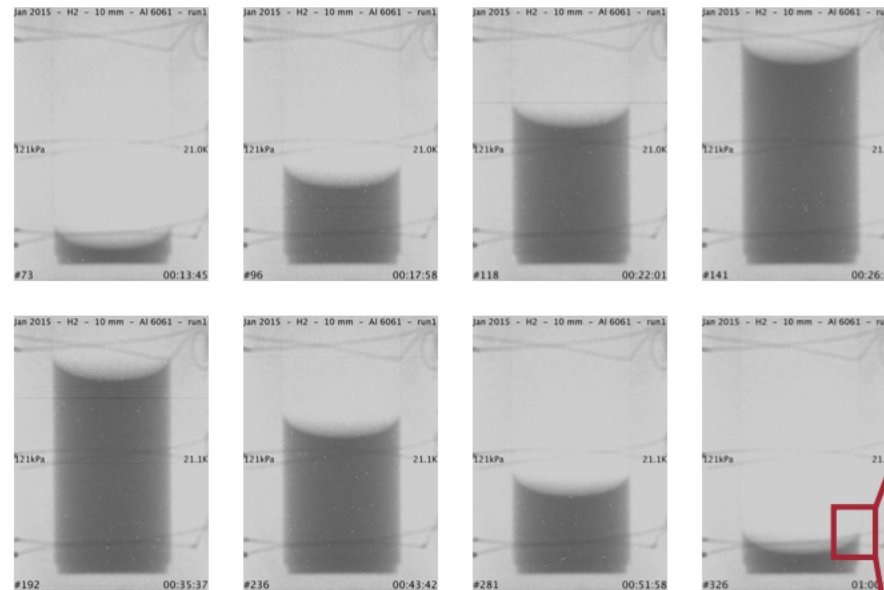
$$h = h \left(\sigma, \mu, \rho, \frac{dP_L}{dx}, \frac{d\sigma}{dx} \right)$$

$$T_i = T_w - \boxed{\dot{m}''} \frac{h_{\text{fg}}}{k} (R - h) \ln \left(\frac{R}{R - h} \right)$$

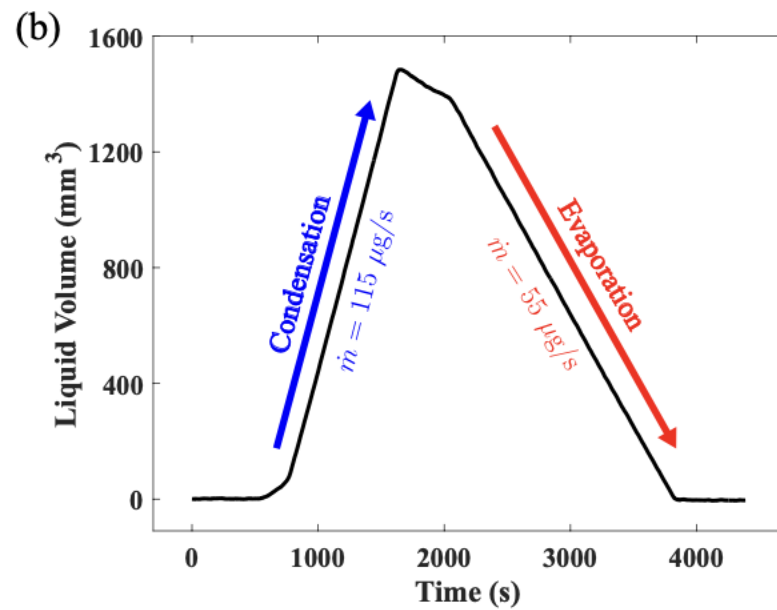




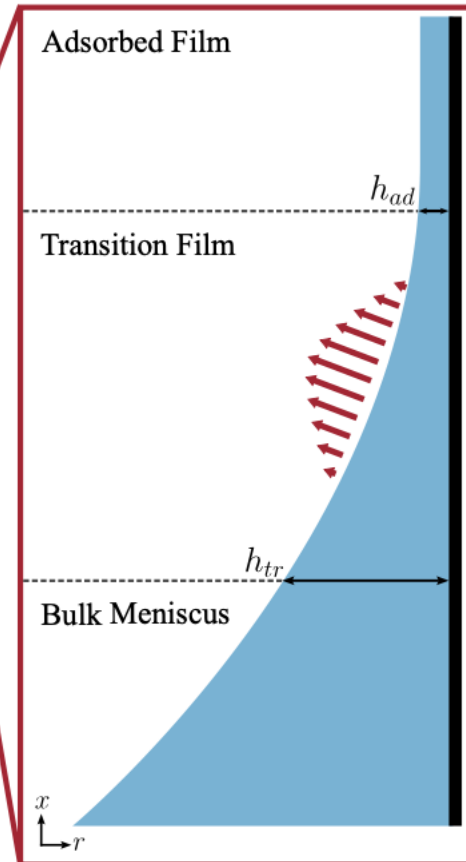
(a)
Condensation

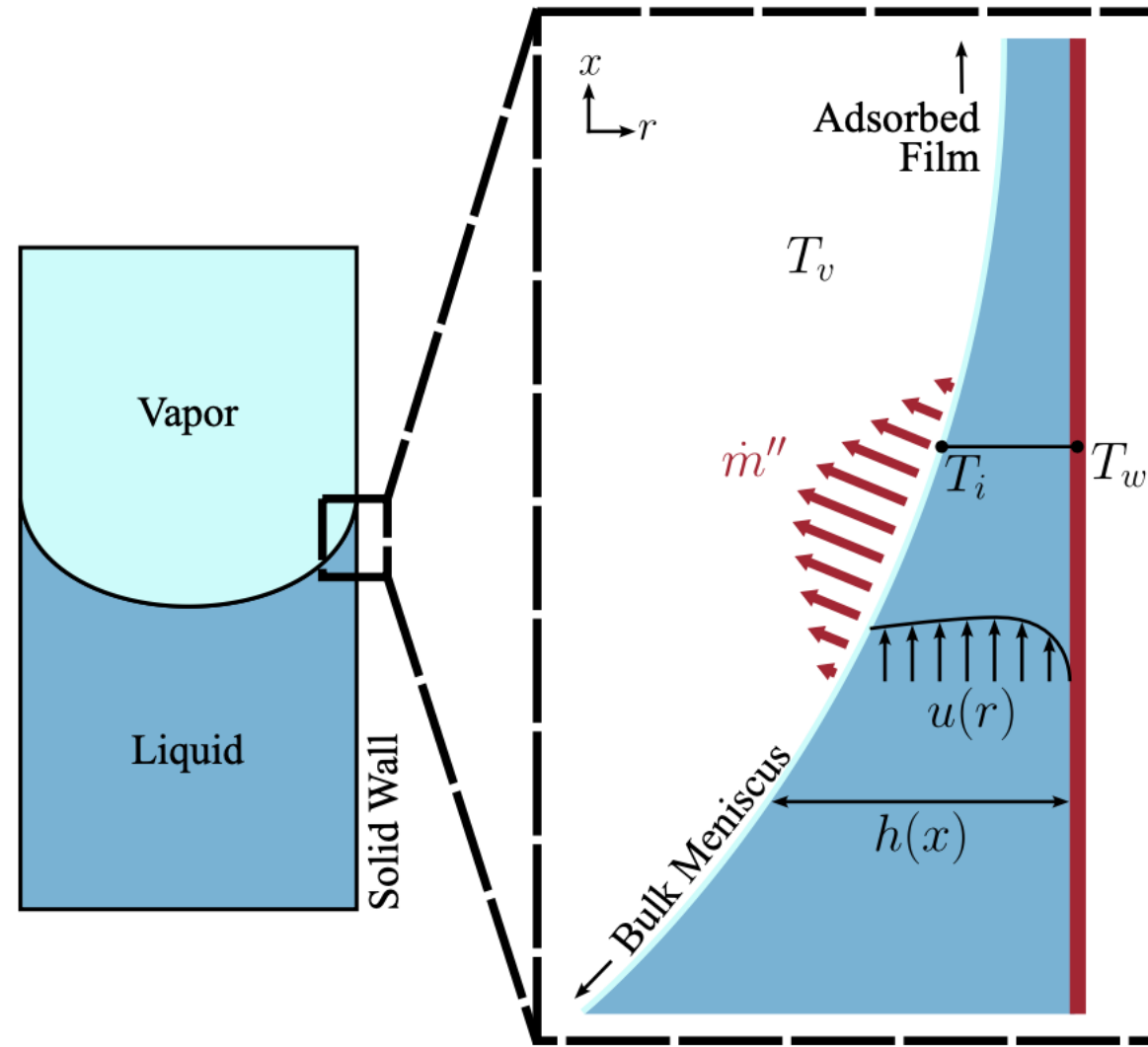


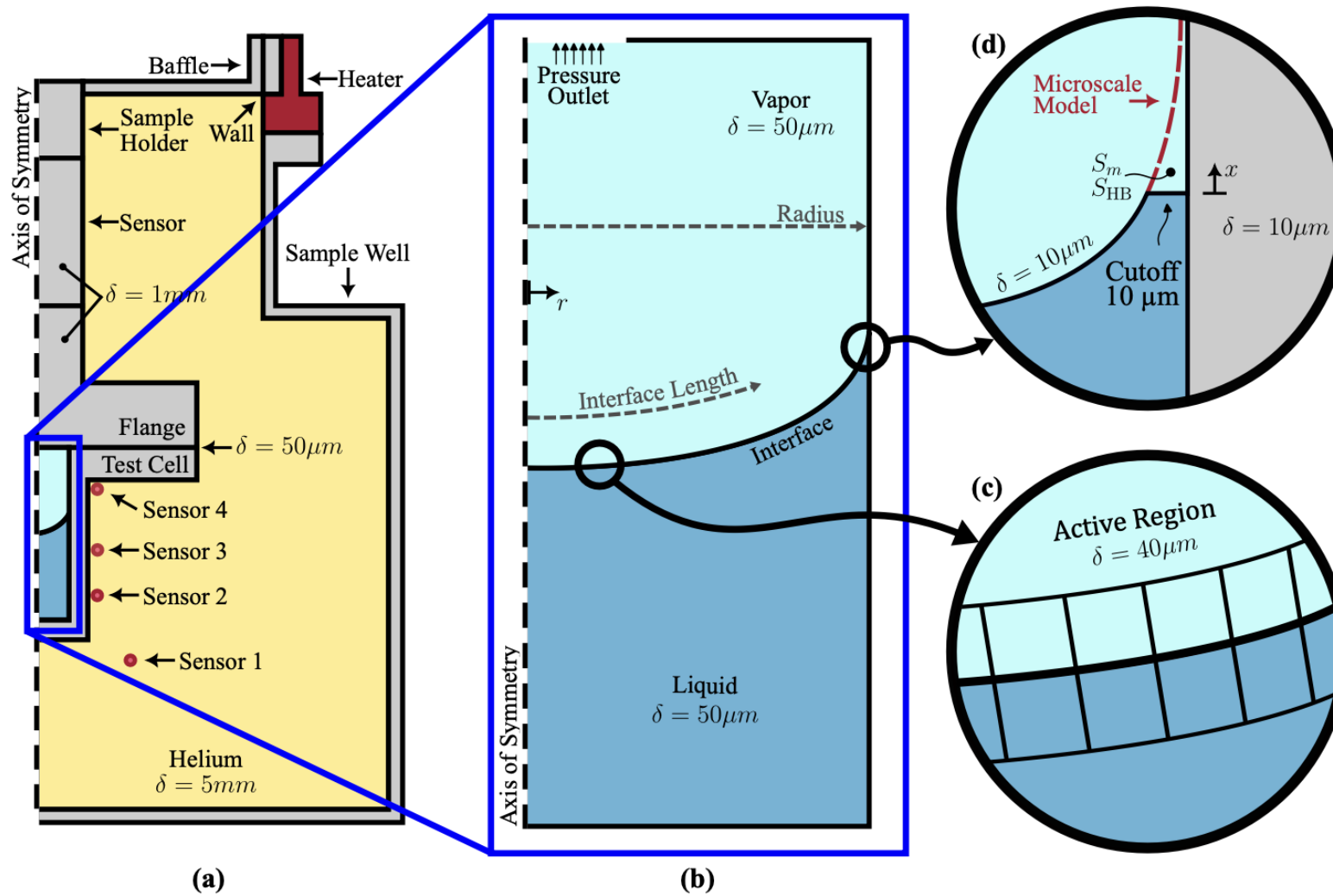
Evaporation

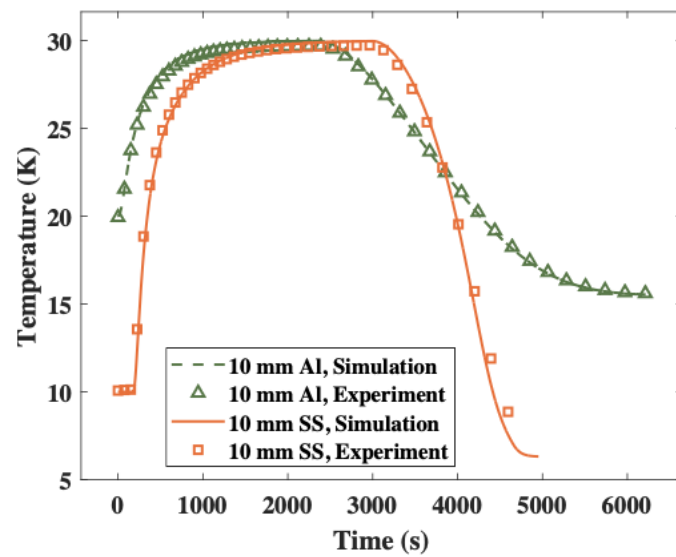


(c) Extended Meniscus

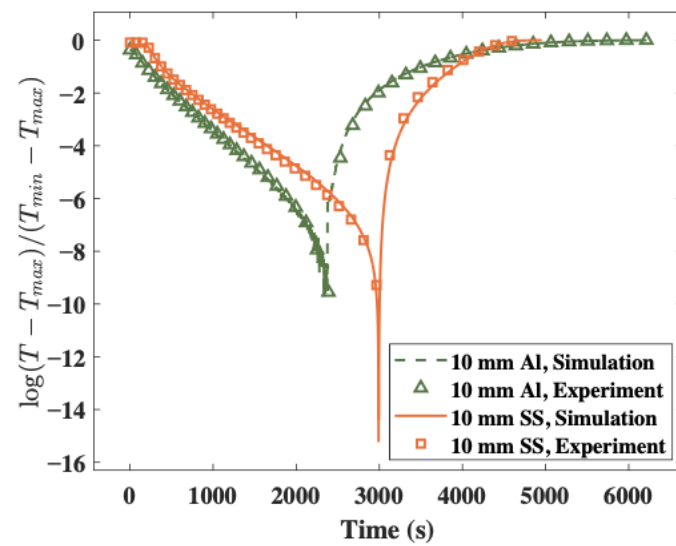




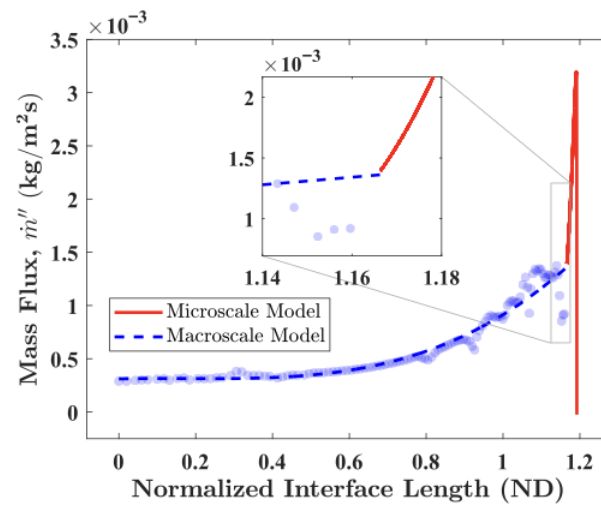




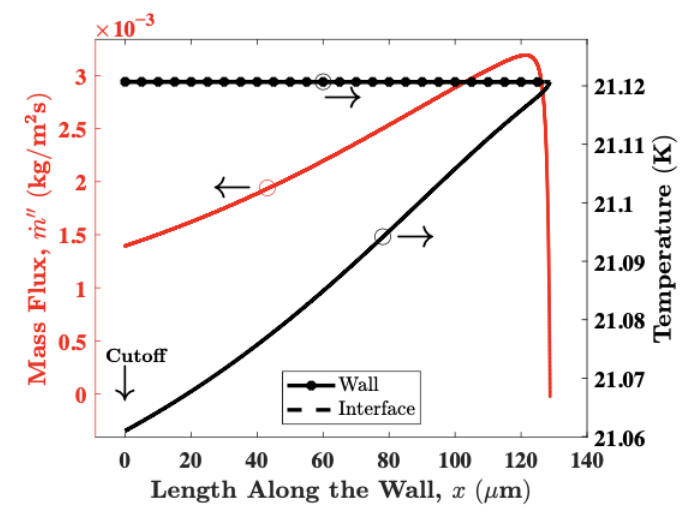
(a)



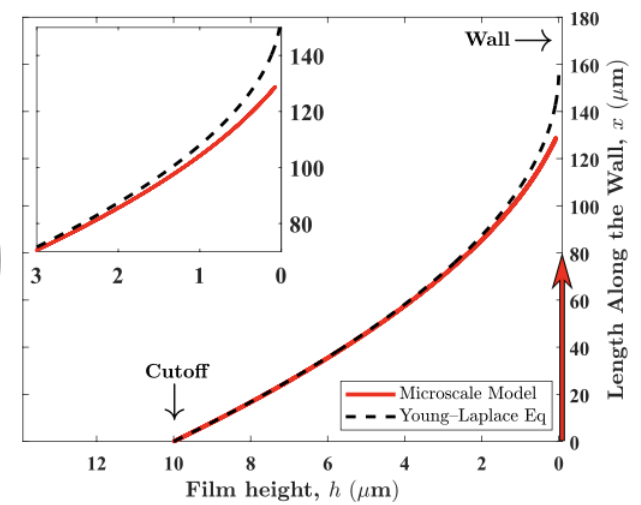
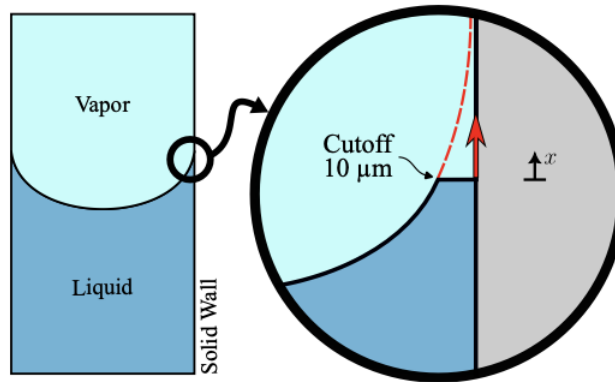
(b)



(a)



(b)



Cryo-Neutron Experiments [35]				Present Study			Bellur et al. [16]		Yasin and Bellur [49]
Test Cell	Run	Pressure (kPa)	$\dot{m}_{exp}$	$\dot{m}_{total}$	$\dot{m}_{bulk}$	$\dot{m}_{tf}$	$\dot{m}_{bulk}$	$\dot{m}_{tf}$	$\dot{m}_{bulk}$
tc2 (10 mm SS 316)	1	120.9	16.43	16.40	11.94	4.46	14.17	2.25	16.93
	2	88.32	17.27	17.20	12.83	4.37	14.72	2.49	17.27
	3	201.96	21.39	21.56	16.19	5.36	18.54	2.84	21.40
tc4 (10 mm Al 6061)	1	121.3	55.50	55.26	46.09	9.17	43.74	11.18	54.67
	2	87.9	55.20	55.25	46.14	9.11	43.44	11.75	54.95
	3	200.05	93.12	92.80	79.02	13.78	73.89	19.22	91.71
	4	226.84	77.31	77.84	63.63	14.21	62.85	14.47	—